

Where semi-analytical ocean-colour retrieval breaks in the PACE era

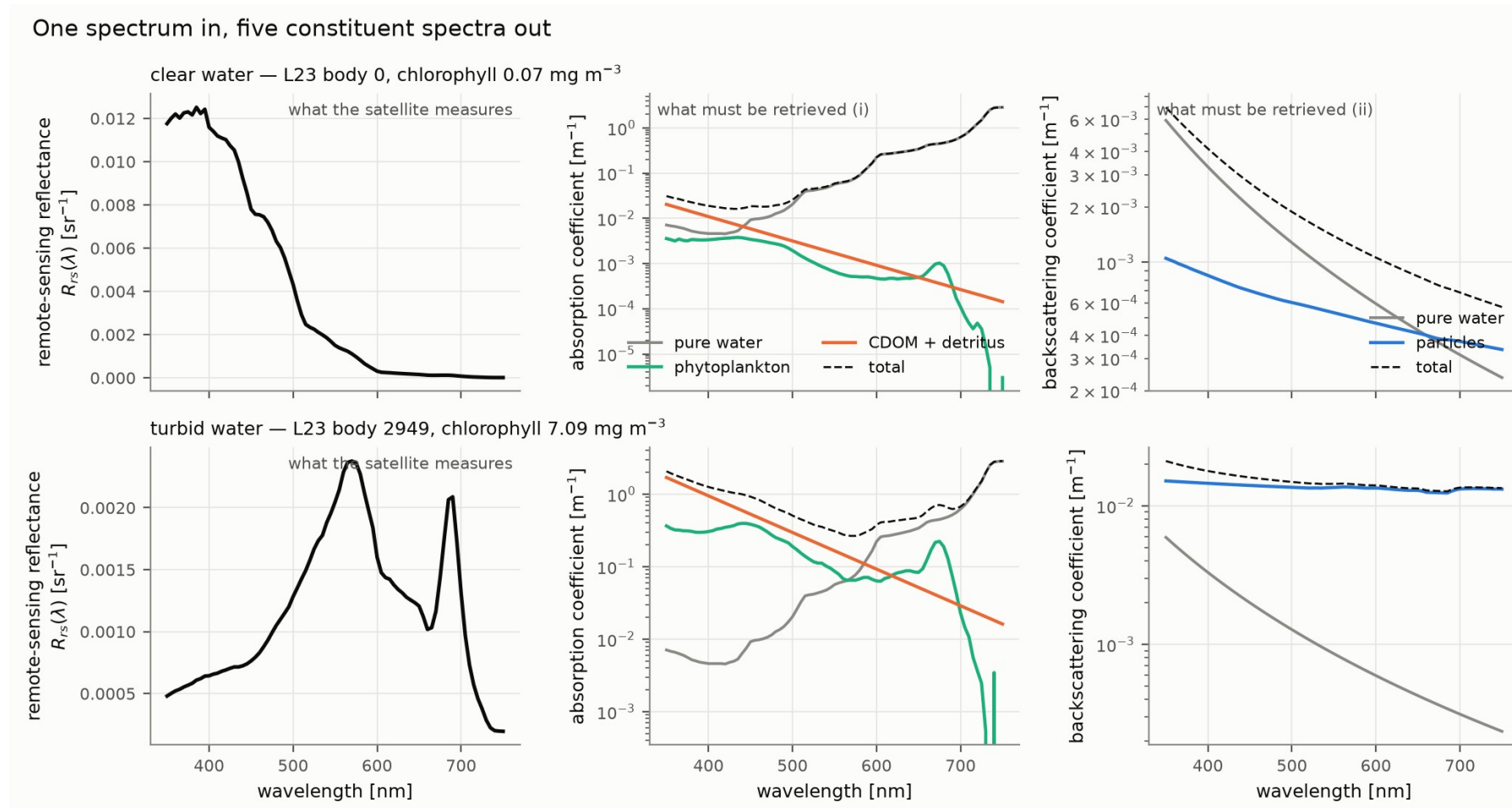
Separating forward-model error from parameterization error

Semi-analytical retrieval of ocean inherent optical properties fails in two separable ways. Forward-model (physics) error is identifiable and removable with a sufficiently accurate differentiable radiative-transfer model; once it is removed, the residual error is parameterization error, concentrated in the split of absorption between phytoplankton and CDOM + detritus.

Candidate: Claude (Anthropic) · Advisor and chair: J. Xavier Prochaska · Committee: Jessica Werk, John O'Meara

Qualifying examination, UC Santa Cruz, 21 September 2026

What the satellite measures: one reflectance spectrum per pixel



Remote-sensing reflectance $R_{rs}(\lambda)$, of order 10⁻³ sr⁻¹; PACE samples it every ~5 nm from 340 to 890 nm. Everything of interest is on the right.

What must be retrieved: five constituent spectra

Absorption $a(\lambda)$ [m^{-1}]

- pure water — known
- phytoplankton a_{ph} — the chlorophyll signal
- CDOM + detritus a_{dg} — dissolved and non-algal, nearly exponential in wavelength
- Both constituents absorb most strongly in the blue

Backscattering $b_b(\lambda)$ [m^{-1}]

- pure water — known
- particles b_{bp} — sets the overall brightness
- One scalar phase-function ratio B_p enters the physics

The problem

- The sum a is set by the level of R_{rs} and is well constrained
- The split a_{ph} vs a_{dg} is set only by spectral shape
- Three to six numbers per pixel must be pulled from one spectrum

How the field solves it: a frozen physics plus a parameterization

- **Gordon (1988): the semi-analytical shortcut**
 - Sub-surface reflectance $\approx g_1u + g_2u^2$, $u = b_b/(a+b_b)$
 - Two coefficients, fitted once to a Monte Carlo ensemble, used by every retrieval since
- **GIOP (Werdell+2013), GSM (Maritorena+2002)**
 - exponential a_{dg} , fixed-shape a_{ph} , power-law b_{bp} ; least squares
- **QAA (Lee+2002): algebraic inversion instead of a fit**
- **BING (Prochaska & Frouin 2025) — inherited, not a thesis contribution**
 - Same relation, same parameterizations, Bayesian posterior by MCMC
 - Every retrieved quantity carries an uncertainty; models compared by evidence
- **What this thesis adds to BING (2026)**
 - a selectable radiative-transfer backend, free B_p , viewing geometry, inelastic terms fixed and validated, three turbid-water b_{bp} models

The claim, in the language of component separation

Physics error is removable; what remains is the parameterization

Two spectral components whose SUM is well determined have a SPLIT that is not.

Before the degeneracy can be blamed on the parameterization, the forward model must be shown accurate enough not to be the cause.

Structurally the same problem as separating dust attenuation from stellar age in an SED fit — named once, then dropped.

Semi-analytical retrieval of ocean inherent optical properties fails in two separable ways. Forward-model (physics) error is identifiable and removable with a sufficiently accurate differentiable radiative-transfer model; once it is removed, the residual error is parameterization error, concentrated in the split of absorption between phytoplankton and CDOM + detritus.

Four projects, six chapters, and who did what

- Ch 2 retrieve-or-bust — the differentiable forward model (robust.rt)
- Ch 3 IOptics — the separation: RT ladder + benchmark on L23, PANGAEA, GLORIA
- Ch 4 EPFT-UP — MOANA reimplemented: parameterization transfer one level up
- Ch 5 PAB — PACE vs BGC-Argo at mission scale (14,610 matchups)
- Ch 1 and 6: introduction and conclusions
- Excluded by the advisor: the bbp700 validation (a collaborator's), the RoB inversion, information content as frame
- The candidate did not choose the problems; the advisor set them and holds a veto
- The candidate wrote the code, ran the analyses, drafted every report
- The advisor wrote the questions, judged the answers, ran every git command
- About 530 dated session logs across four repositories since 2026-06-15
- Every number in this talk traces to one of them; every citation resolves to a path, branch and commit

Chapter 2 — a forward model accurate enough to be blamed for nothing

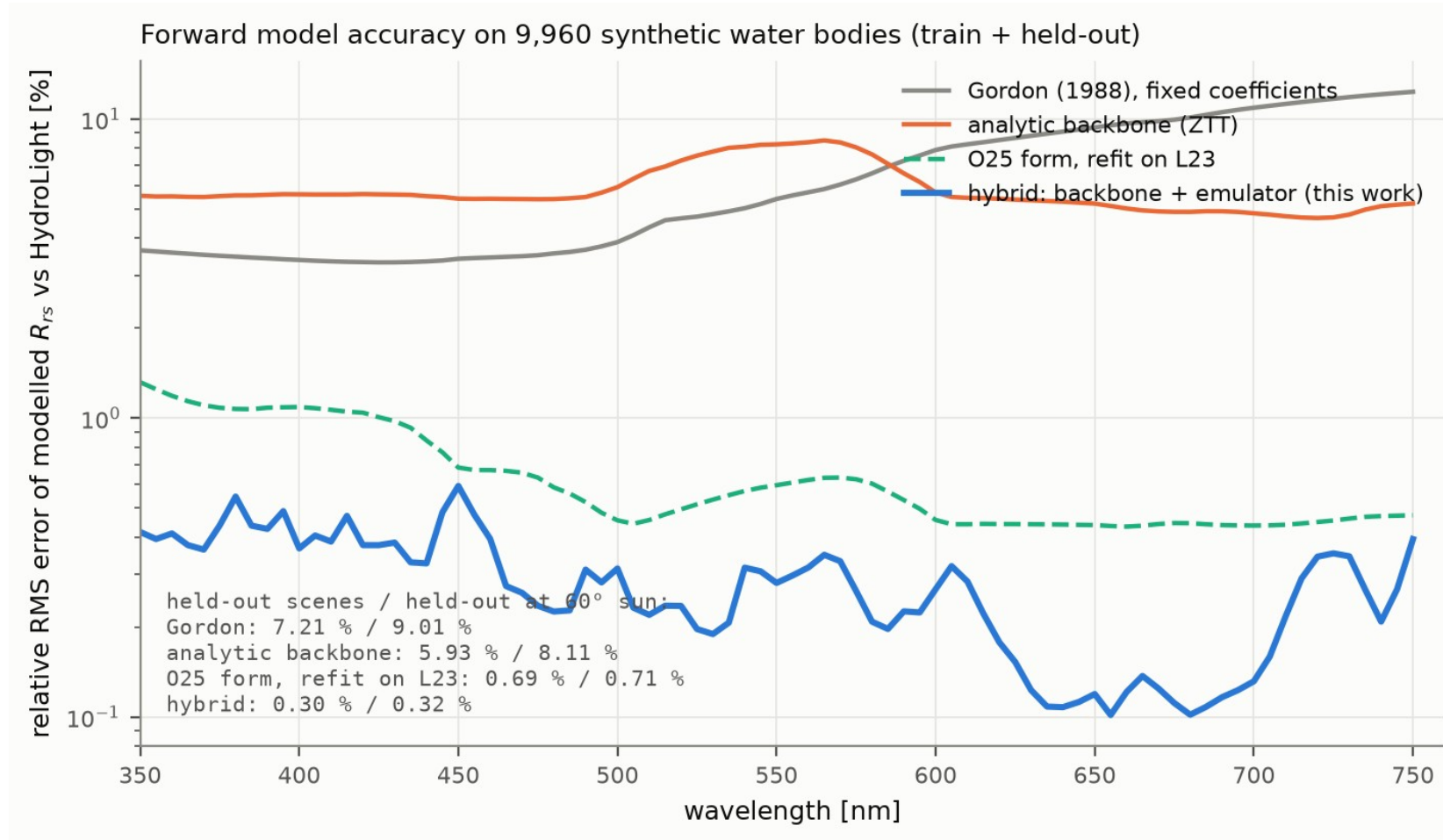
- **Hybrid: analytic backbone $\times (1 + \delta)$**
- Backbone: Zaneveld / Twardowski–Tonizzo (ZTT) analytic model
- δ : a 417-parameter neural correction trained on HydroLight
- Written in JAX — differentiable end to end, callable inside an MCMC likelihood
- **Training set: L23**
- 3,320 HydroLight water bodies $\times$ 81 wavelengths (350–750 nm) $\times$ 3 solar zeniths (0°, 30°, 60°)
- Held out: 20 % of bodies; and a whole zenith

0.30 %

relative RMS error of modelled R_{rs} against HydroLight on held-out water bodies

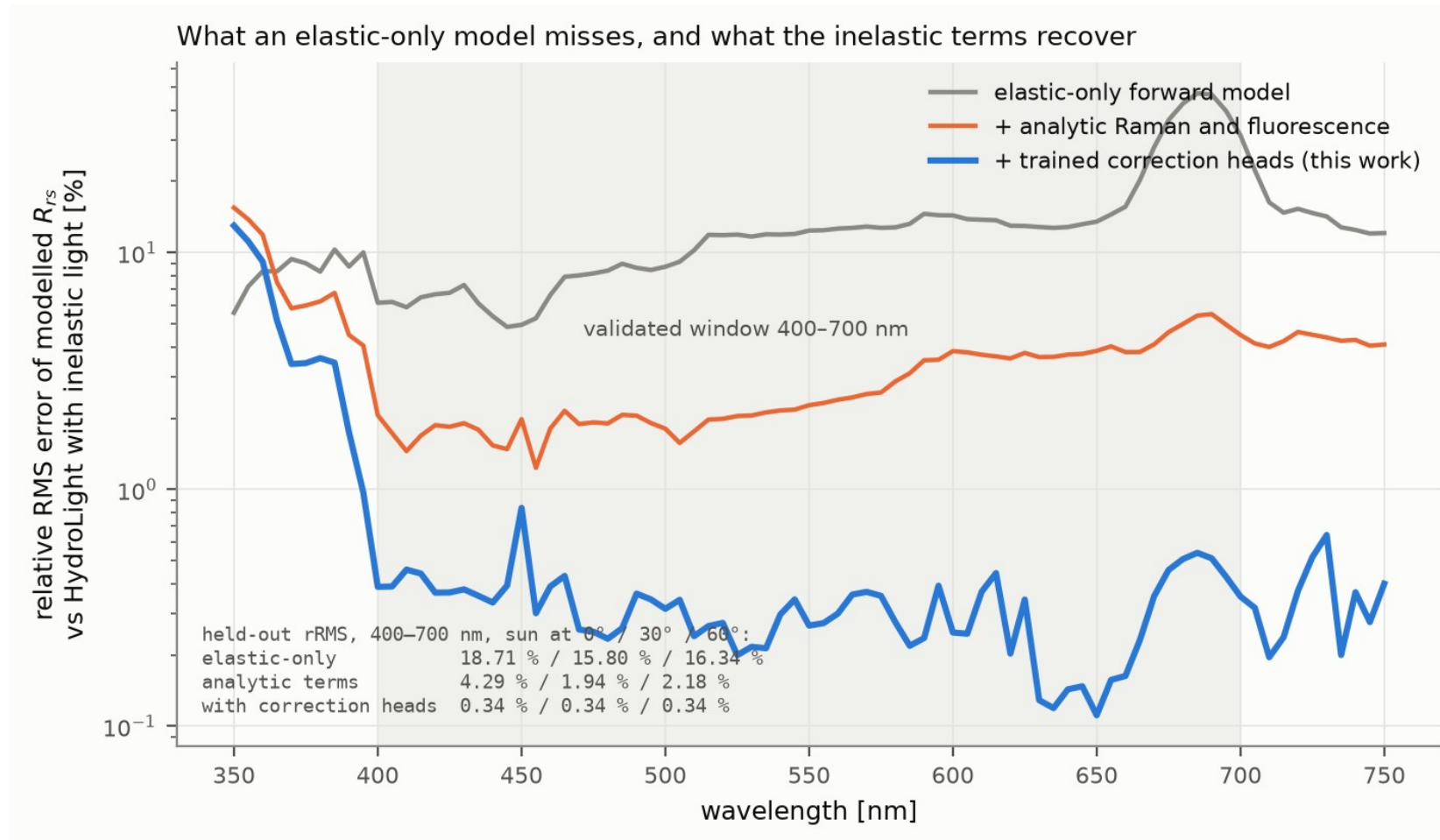
- 2.3× better than the best published semi-analytical form refit on the same data
- 24× better than the Gordon relation
- exact gradients: agree with finite differences to 5×10^{-9}

Forward-model accuracy: from 7 % (Gordon) to 0.3 % (hybrid)



The forward-model error term can be driven small enough that whatever retrieval error remains is not the physics.

Inelastic light is 15–20 % of the red-end signal an elastic model cannot fit



Raman scattering and chlorophyll fluorescence re-emit light; with analytic terms the error is 2–4 %, with trained correction heads 0.34 %.

The forward model owns its edges

- **Solar zenith: trained at 0°, 30°, 60°; nadir view only**
- Held-out zenith: 4.7–12.2 % (elastic), –74 % on the Raman head (inelastic)
- **Phase function: L23 spans only 1.7× in B_p — generalisation untested**
- **CDOM fluorescence: implemented, unvalidated — no HydroLight truth contains it**
- **Off-nadir saturation bug found 2026-09-15 while PAB 2.0 exercised it; fixed**
- Every sweep in this talk states which of these it is inside

- **What the forward model is NOT**

- It is not a retrieval. The retrieve-or-bust inversion does not exist yet
- "nothing on this site may be read as evidence that IOPs can be retrieved" — RoB docs
- The inversion belongs to a seven-person collaboration and is not anticipated here

- **What this thesis claims**

- The forward model, its validation, and its use inside BING as a physics backend

Extensions to BING made during the thesis

- **Radiative-transfer backend**

- $\text{rt_backend} \in \{\text{gordon}, \text{robust_ztt}, \text{robust_hybrid}\}$; per-observation viewing geometry; free B_p as a sixth parameter

- **Inelastic terms fixed and anchored to L23**

- a missing $1/\pi$ in the fluorescence normalisation and a flat E_d ratio in Raman: amplitudes had been $\sim\pi$ too large

- **CDOM fluorescence path**

- $a_{\text{cdom}} = 0.8 \times a_{\text{dg}}$ feeds the Hawes (1992) kernel — a fixed fraction, stated wherever used

- **Three turbid-water b_{bp} models**

- Pow2 (two power laws), Pow2Flat, PowFlex — built to answer GLORIA (slide 25)

- **Where the attribution line falls**

- BING as of the 2025 paper is inherited; the 2026 additions above are the thesis's, as plumbing for the new physics and as parameterization work

- **Status**

- on BING branches develop and rob_cdom; the advisor merges after submission

Chapter 3 — the RT ladder: one parameterization, five radiative transfers

rung	forward model	what changes
1	elastic, analytic (ZTT)	the analytic backbone alone
2	elastic, hybrid	+ the trained emulator correction
3	+ Raman scattering	first inelastic process
4	+ chlorophyll fluorescence	second inelastic process (685 nm)
5	+ CDOM fluorescence	third, unvalidated; $a_{\text{cdom}} = 0.8 a_{\text{dg}}$

Same parameterization (exp. a_{dg} + Bricaud a_{ph} + power-law b_{bp} + B_{p}), same priors, same noise, same spectra, full MCMC. Three arms: 3,320 L23 bodies with truth (inelastic realization), 97 PANGAEA in-situ spectra with truth, 99 real PACE pixels without.

Why five rungs: each step switches on one physical process

- **Rung 1 → 2: analytic backbone → hybrid**

- the trained correction changes R_{rs} by 1.4–3.7 %; the control for everything the emulator does

- **Rung 2 → 3: + Raman scattering**

- inelastic light redistributed from blue to red; changes modelled R_{rs} by ~2 % of its maximum in the blue

- **Rung 3 → 4: + chlorophyll fluorescence**

- a peak at 685 nm, ~35 % of R_{rs} there in L23

- **Rung 4 → 5: + CDOM fluorescence**

- a broad hump peaking near 515 nm; the unvalidated rung, with $a_{cdm} = 0.8 a_{dg}$

- **Equal parameter count on every rung**

- so model selection between rungs is a pure likelihood contest

- **One noise model per arm**

- PACE-like on L23, each record's own error (10 % floor) on PANGAEA, per-pixel R_{rs_unc} on PACE

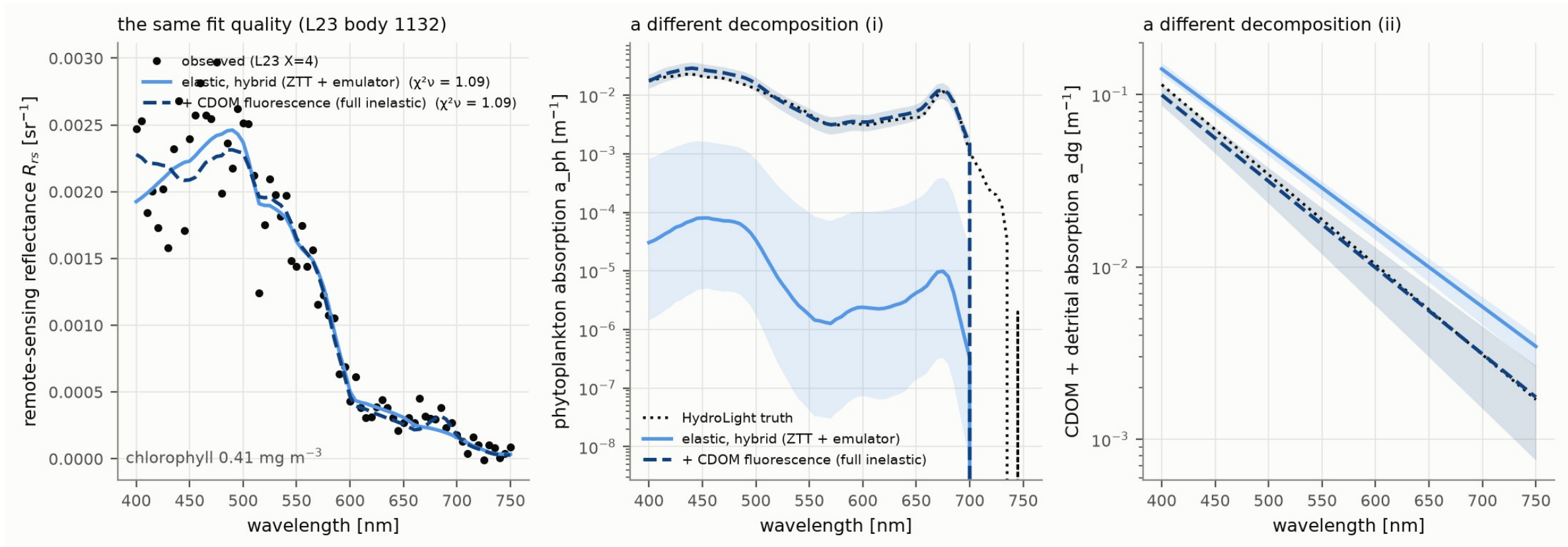
- **Truth only where it exists**

- L23: exact IOPs; PANGAEA: measured a_{ph} , a_{dg} , b_{bp} ; PACE: none — that arm scores closure and model selection only

- **Cost**

- 17,020 MCMC fits for RT-A (171 h on 20 cores), 495 for RT-B (4.8 h)

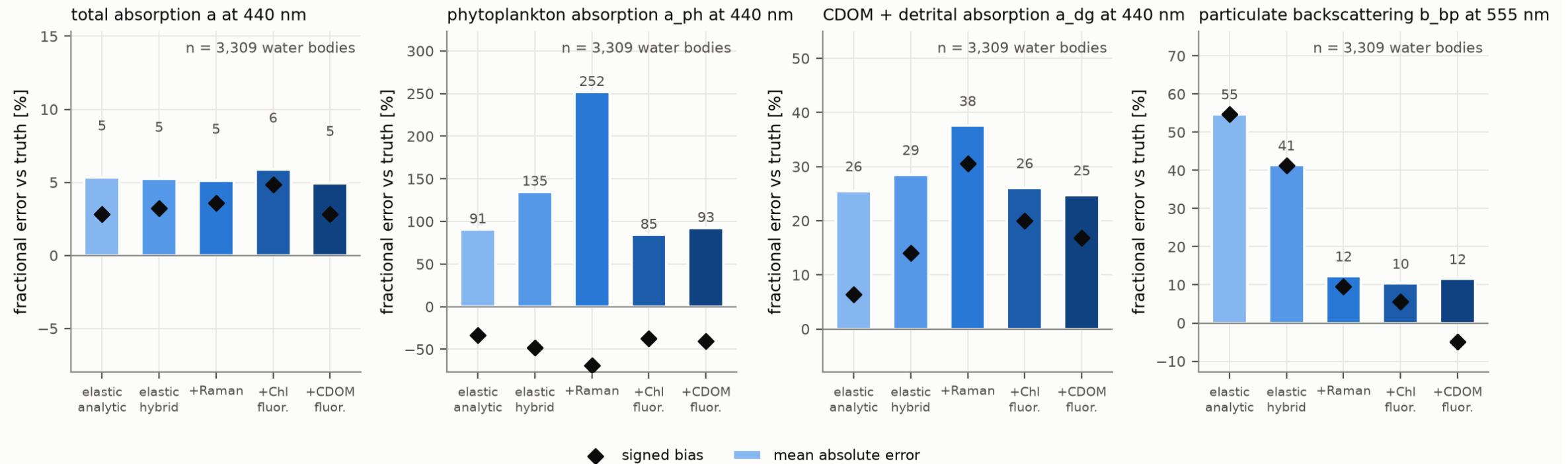
Two forward models, one spectrum: the same fit, a different decomposition



Both fits reach $\chi^2\nu = 1.09$. Phytoplankton absorption differs by orders of magnitude. The fit cannot tell the physics apart.

The centrepiece: down the ladder, error moves only where the physics was the problem

Five forward models, one parameterization, L23 synthetic truth — error moves down the ladder only where the physics was the problem



Total absorption: 5 % everywhere. Particulate backscatter: +55 % → -5 %. Phytoplankton and CDOM absorption: unchanged.

Read it as two findings

+55 % → -5 %

particulate backscatter bias, elastic → full inelastic (L23, 555 nm)

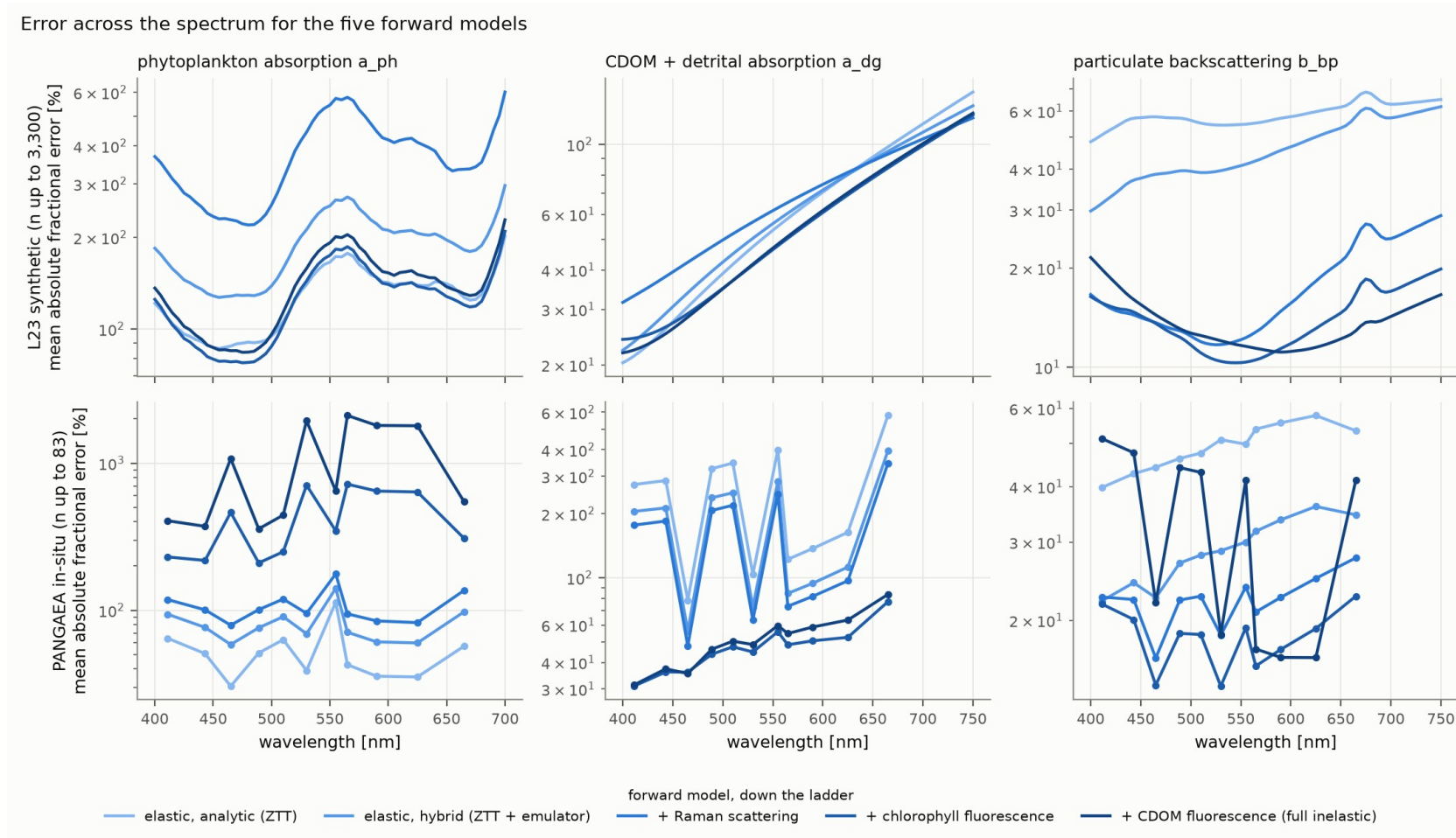
- interval coverage 0.00 → 0.59
- this is a physics failure
- and it is the failure the forward model was built to remove

×2 wrong

phytoplankton absorption under every one of the five physics (MAE 85–252 %, always low)

- a_{dg} 25–38 % high under every physics
- $\chi^2v = 1.07$ –1.14 throughout: the fit is happy
- this is a parameterization failure that no physics fixes

The same ladder across the spectrum, on synthetic and in-situ spectra



On L23 the physics fixes backscatter at every wavelength; on PANGAEA the fluorescence terms move error from CDOM into phytoplankton rather than removing it.

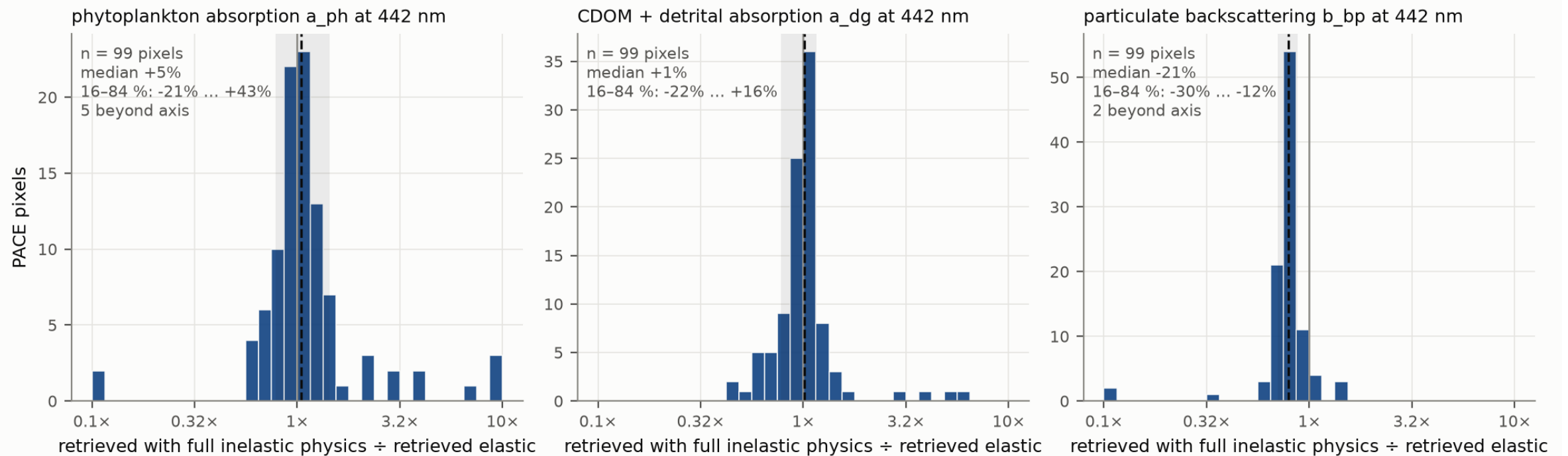
The in-situ arm: 97 PANGAEA spectra with measured a_ph, a_dg, b_bp

quantity	elastic hybrid	+ Chl fluorescence	what moved
a_dg(440) error	213 %	36 %	much better
a_ph(440) error	76 %	218 %	much worse
b_bp(555) error	30 %	19 %	better
χ^2v (median)	1.45	2.10	worse fit under flat 10 % error
fits acceptable	86 %	74 %	thin 6–11-band spectra; B_p fixed

On real spectra the physics shifts error between the two absorption components; it does not remove it. B_p is fixed here (Q52) and the error model is a flat 10 % (Q53).

Real PACE pixels, no truth: how far does the physics move the answer?

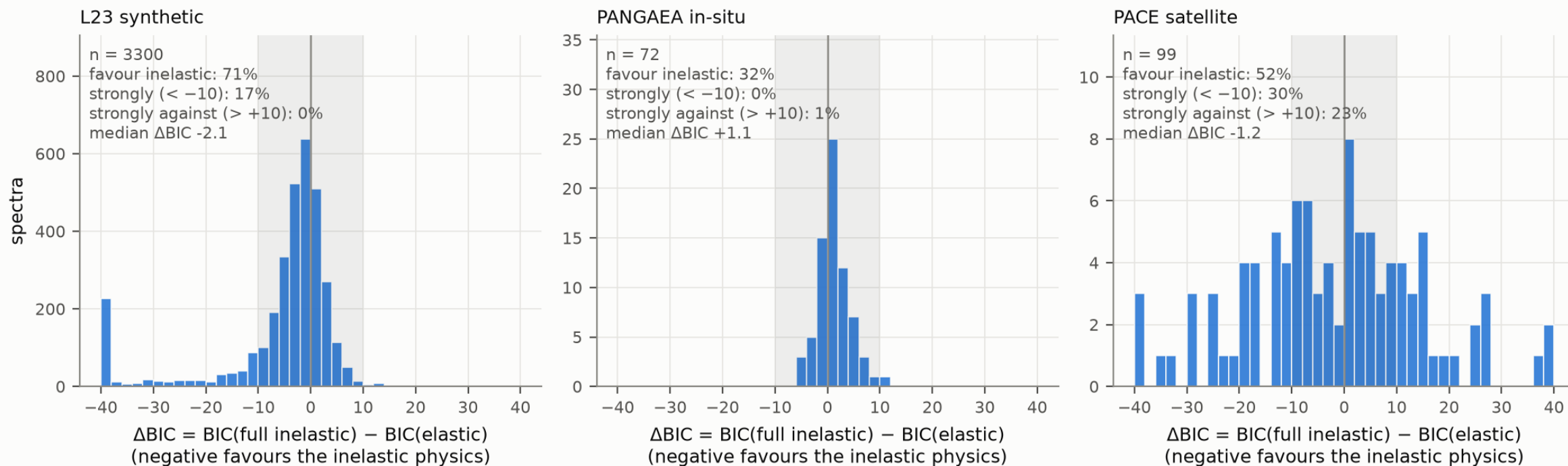
Same parameterization, two forward models, 99 real PACE pixels: how much the physics moves the answer



Full inelastic $\div$ elastic on 99 pixels: backscatter -21 % (16-84 %: -30...-12), a_{ph} +5 %, a_{dg} +1 % — nearly unchanged on average, wide spread.

Would the fit alone have found the physics error? No.

Does the fit itself prefer the fuller physics? Same parameterization, equal parameter count



ΔBIC , equal parameter counts: synthetic spectra prefer the inelastic physics (71 %), sparse in-situ spectra do not (32 %), satellite pixels split (52 %; 30 % strongly, 23 % strongly against).

What the ladder pages say about themselves

- **Held fixed, stated on every page**

- nadir viewing geometry ($\theta_v = 0$, $\Delta\phi = 0$); solar zenith 0° on L23, computed per record elsewhere
- $a_{\text{cdom}} = 0.8 \times a_{\text{dg}}$ drives the CDOM-fluorescence term
- packaged sky irradiance (ed_l23.npz)
- learned inelastic corrections OFF inside BING — the ladder measures the analytic inelastic physics
- hybrid emulator evaluated outside its trained B_p span wherever B_p is free (accepted, Q50c)
- L23 truth lacks CDOM fluorescence and uses a single-Gaussian emission line

- **Consistency**

- 99 PACE pixels reproduce PAB's Gordon-elastic posteriors: correlations 0.94–1.00, derived-Chl ratio 1.05, identical input spectra

- **Completeness**

- L23: 3,309 of 3,320 fitted on every rung (11 red-peaked bodies declined identically); PANGAEA 95 of 97; PACE 99 of 100

- **Cost**

- 17,020 MCMC fits, 171 h on 20 cores for RT-A; 4.8 h for RT-B

- **Excluded from the leaderboard by design**

- five rungs of one algorithm are not five algorithms

The 99 PACE pixels are PAB's: the two fitters agree to within the physics swap

shared parameter	scale	correlation of posterior medians	median (ours – PAB)	16–84 % span
A_dg (CDOM amplitude)	log10	0.987	–0.010	–0.048 ... +0.037
S_dg (CDOM slope)	linear	0.939	–0.0007	–0.0014 ... 0.0000
A_ph (phytoplankton amplitude)	log10	0.965	+0.022	–0.027 ... +0.134
B_nw (backscatter amplitude)	log10	0.998	+0.048	+0.034 ... +0.065
β (backscatter slope)	linear	0.935	–0.078	–0.367 ... +0.014
derived chlorophyll (ratio)	ratio	0.965 (log)	1.053	0.94 ... 1.36

Same spectra to the last bit ($\max |\Delta R_{rs}| = 0$ on all 136 bands). The +0.05 dex in B_{nw} is the elastic-model swap (hybrid sits above Gordon in R_{rs}) plus the free B_p . The PACE arm stands on the data PAB published from.

The benchmark: one physics, any parameterization, uniform scoring

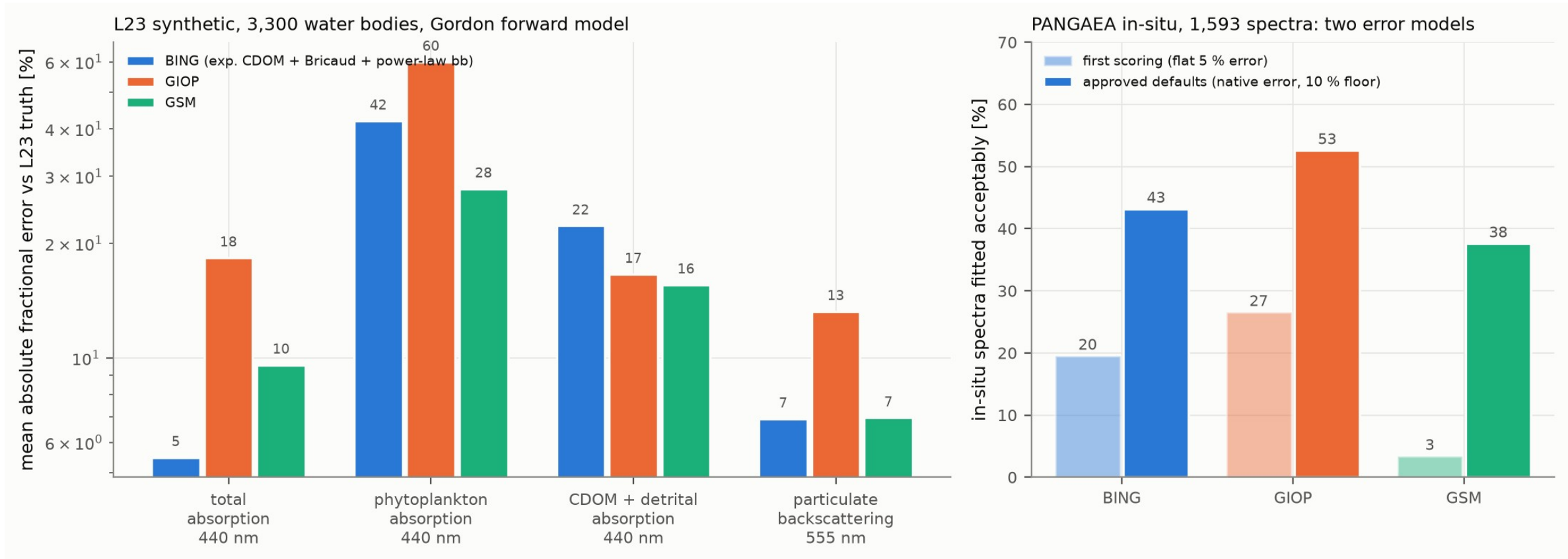
- **IOptics: a framework, not a script**
- adapters for L23, PANGAEA (Valente+2022), GLORIA (Lehmann+2023), PACE
- algorithms register with one call: BING's expb_pow, GIOP, GSM, the turbid trio, the five RT rungs
- χ^2 and full MCMC paths; provenance-stamped report pages; a cross-sweep leaderboard
- **Scoring**
- log-space accuracy and bias (Erickson+2023 convention), 68/95 % coverage, Δ BIC, paired-bootstrap head-to-head that can say "indistinguishable"
- 535 tests; sphinx -W clean

3,320 + 1,593 + 7,572

water bodies with truth: L23 synthetic, PANGAEA in-situ, GLORIA coastal/inland

- every page carries the code and config versions that produced it
- ioptics.readthedocs.io

Three parameterizations under one physics: totals agree, the split does not



On L23: total absorption 5–18 %, phytoplankton 28–60 %, CDOM 16–22 %. On PANGAEA: half the apparent failure was the error model, not the algorithms (20/27/3 % → 43/53/38 %).

The in-situ failure rate was a scoring artefact

20 / 27 / 3 %

PANGAEA spectra fitted acceptably under a flat 5 % error model and post-hoc scope (BING / GIOP / GSM)

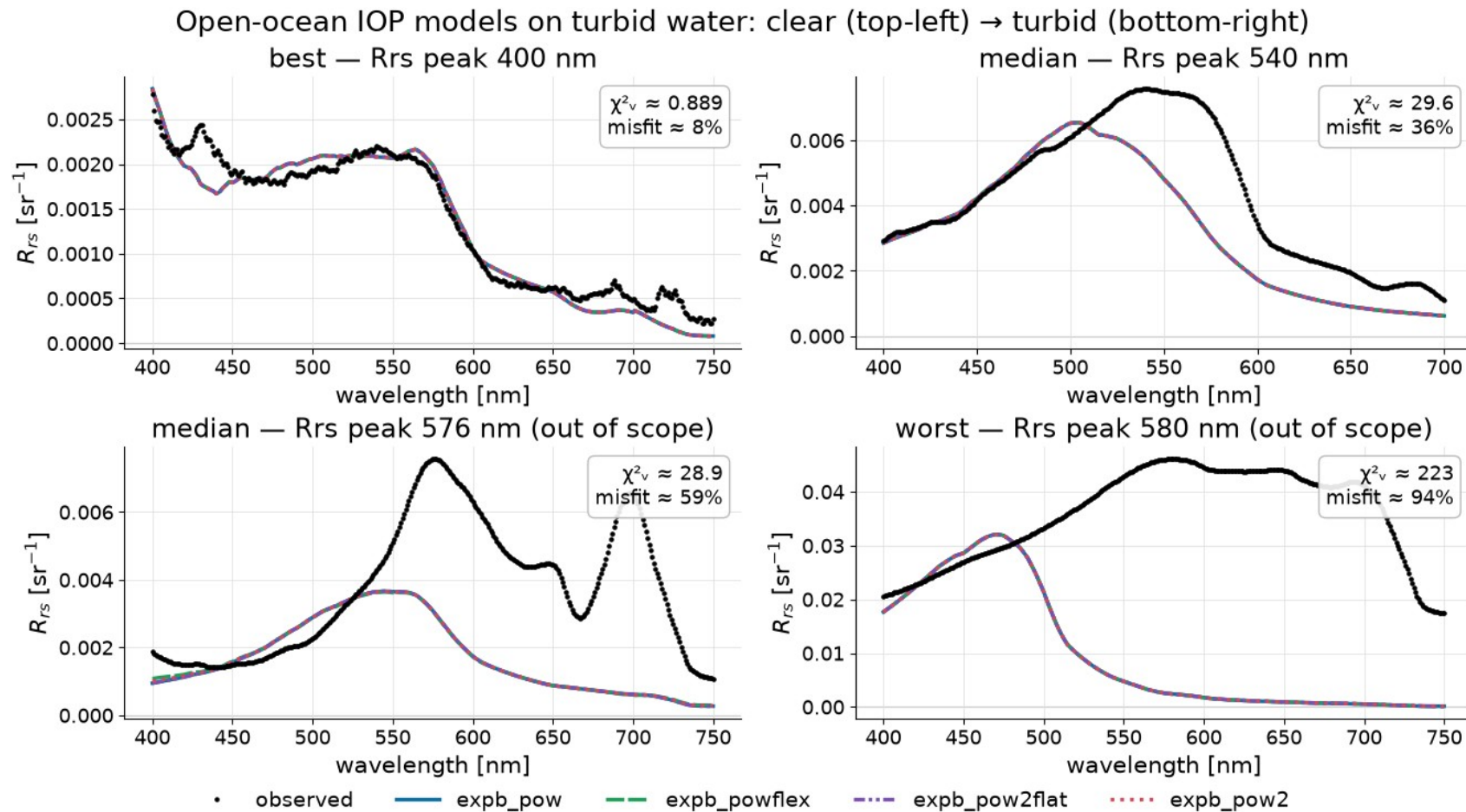
- converged-fit misfit was 6.7–13 %, against L23's 3–5 %
- the compilation quotes no per-band uncertainty

43 / 53 / 38 %

the same spectra under each record's own error with a 10 % floor and a pre-fit scope rule

- 89–93 % valid under the field's own rule (33 % misfit over 400–600 nm)
- the residue tracks instrument era and contributor, not algorithm

Turbid water: four backscatter parameterizations, one identical failing fit



GLORIA: power-law b_{bp} cannot supply the red backscatter ($0.2\text{--}0.4\text{ m}^{-1}$ needed, 0.013 fitted); three richer models return the same fit (χ^2_v $0.460/0.463/0.462/0.460$). The forward model is the remaining suspect.

Least squares vs MCMC on the full L23 population (3,320 bodies)

a(440), BING parameterization	least squares (χ^2)	MCMC posterior
error (fractional MAE)	9.9 %	6.8 %
68 % interval coverage (nominal 0.68)	0.46 — over-confident	0.69 — calibrated
fits that succeeded	3,271 (33 failed, GIOP)	3,304 ok, 16 out of scope, 0 failed
what the interval says about a _{dg} / a _{ph}	over-covers (0.94 / 0.97)	over-confident (0.63 / 0.58)

The sampler is both more accurate and honest about total absorption; on the decomposition neither method's interval is right, which is the degeneracy again.

What the ladder and the benchmark say together

Chapter 3 in four lines

Fix the parameterization, vary the physics: backscatter error moves, absorption split does not.

Fix the physics, vary the parameterization: absorption split moves by factors, totals hardly at all.

Push the backscatter parameterization on turbid water: nothing changes; the forward model is the suspect, and the ladder is the test.

Physics error is removable. What remains is the parameterization, and it lives in a_{ph} versus a_{dg} .

Chapter 4 — the same question one level up: MOANA and EPFT-UP

- **MOANA (Lange+2020): NASA's operational PACE algorithm**
- abundance of Prochlorococcus, Synechococcus and picoeukaryotes from hyperspectral R_{rs} and SST
- a principal-component regression trained on one cruise (AMT24)
- a parameterization whose retrieved quantity is a cell count, not an IOP
- **EPFT-UP: empirical PFTs with uncertainty and provenance**
- from-scratch reimplementation with QC flags in place of silent clipping
- every reported number regenerated by a script; vendored, checksum-pinned tables

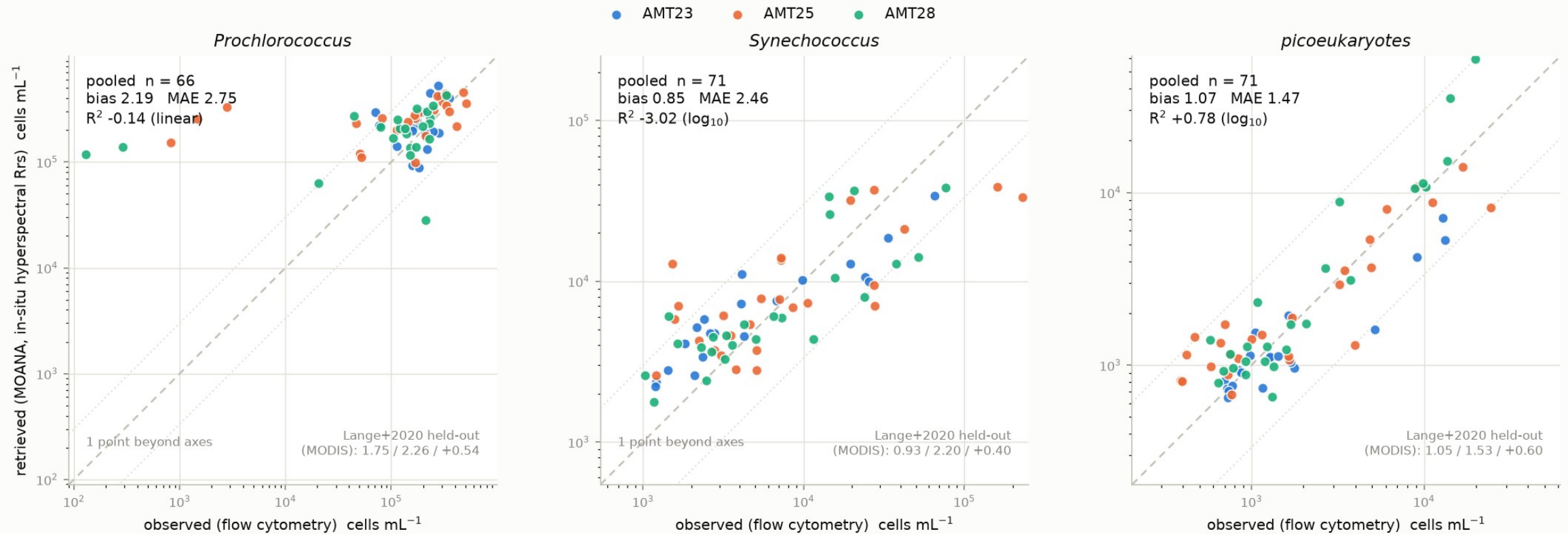
35 / 35

report-printed numbers reproduced at printed precision after the code moved repositories; all six figures pixel-identical

- the only chapter with a passing re-derivation audit
- v0.1.0, Zenodo DOI 10.5281/zenodo.22797970

Held out for the first time: picoeukaryotes transfer, the cyanobacteria do not

Held-out skill: the published MOANA coefficients on AMT23/25/28 in-situ hyperspectral Rrs (dashed 1:1, dotted $\pm 3\times$)



Published MOANA coefficients on independent hyperspectral in-situ spectra (AMT23/25/28): picoeukaryotes bias 1.07, R^2 0.78; Prochlorococcus bias 2.19, $R^2 < 0$; Synechococcus $R^2 -3.0$.

Three findings about the operational product

100,000 px

granule test: the shipping product uses the operational, not the published, coefficient assignment

- Synechococcus median $\Delta\log_{10}$ -0.005 (operational) vs $+0.082$ (published)

18 %

of Prochlorococcus retrievals in the product are exactly zero — clipped, not measured

- 8.9 % for Synechococcus; land encoded inside the valid range

|cos| 0.999

NASA's PCA basis recovered from the raw training cruise on the leading component (0.998 on the second)

- Lange Table 1 reproduced in the CTD-only configuration

Status and the one blocker

- **Exists**
 - report rev. 4; re-derivation audit (PASS); README; 16-page documentation site; CI; CITATION.cff; v0.1.0 with DOI
- **Blocked**
 - the full retrain on AMT24 needs the underway flow-cytometry counts, held at Plymouth Marine Laboratory and in no public archive
 - two requests have not yet produced them; the wording looks like the problem, not a refusal
- **Chapter written to stand without the retrain**
 - the audit and the transfer test are the evidence; the retrain is an upgrade, not a rescue
- **How it fits the claim**
 - a parameterization tuned on one cruise fails to transfer for two of three groups —
 - the same shape as a_ph vs a_dg: the model is right where the training sampled the signal broadly and wrong where it did not
- **Out of scope**
 - the diatom work on a collaborator's algorithm, on a branch of the same repository

Chapter 5 — PACE against BGC-Argo at mission scale

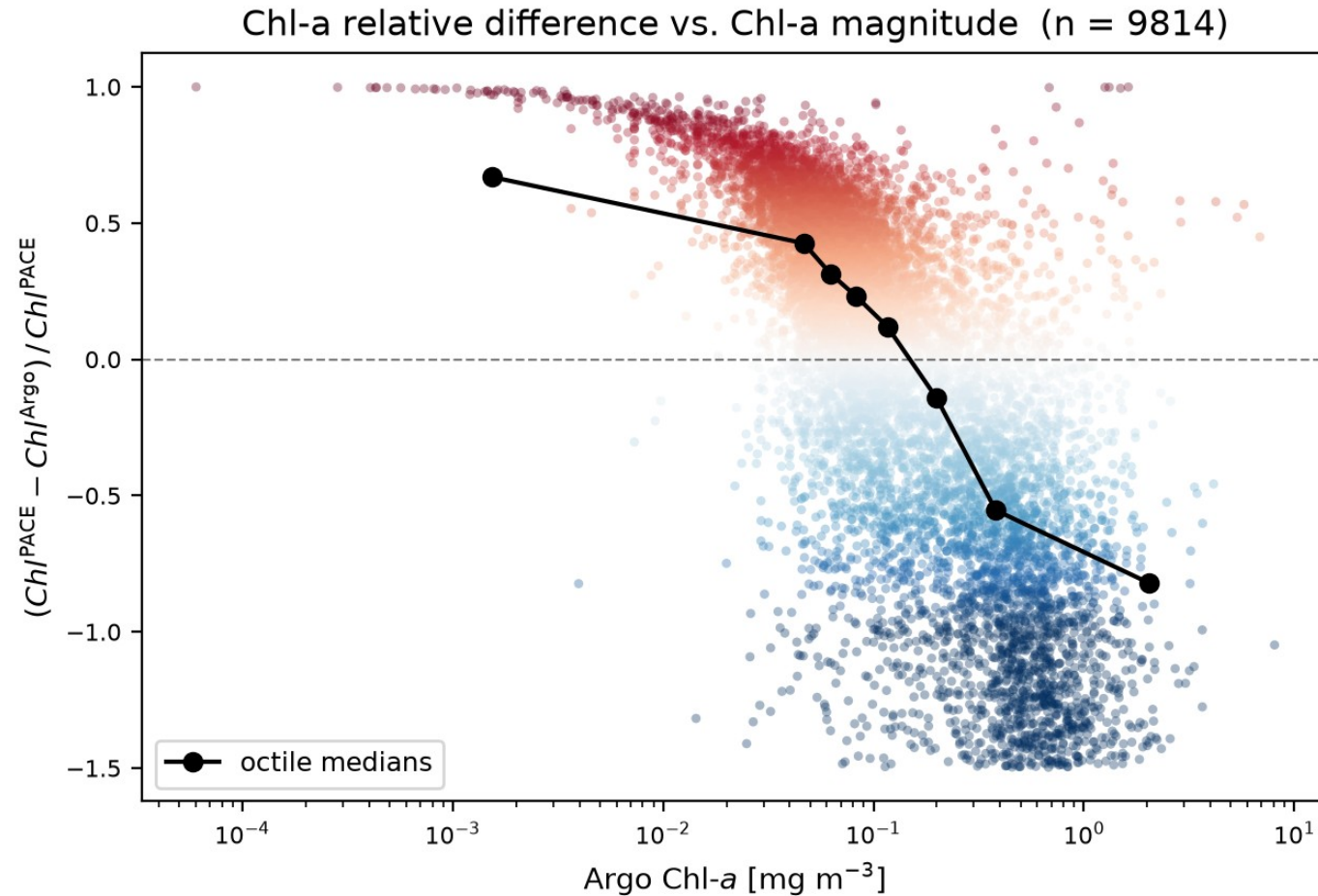
- **PAB: every PACE L2 pixel within 5 km and 24 h of a float surfacing**
- BING fitted to the pixel; float chlorophyll, backscatter and CDOM fluorescence as independent truth
- **Production run (pab_version 1.0, 2026-08-20, Nautilus cluster)**
- 881 floats · 54,031 profiles · 67,435 granules · 14,610 matchups · 14,609 fits
- median separation 0.80 km, median Δt 10.2 h
- published as a SQLite database on s3://pab; 18 GB of chains backed up

9,814

matchups with valid chlorophyll in the Chl-a report

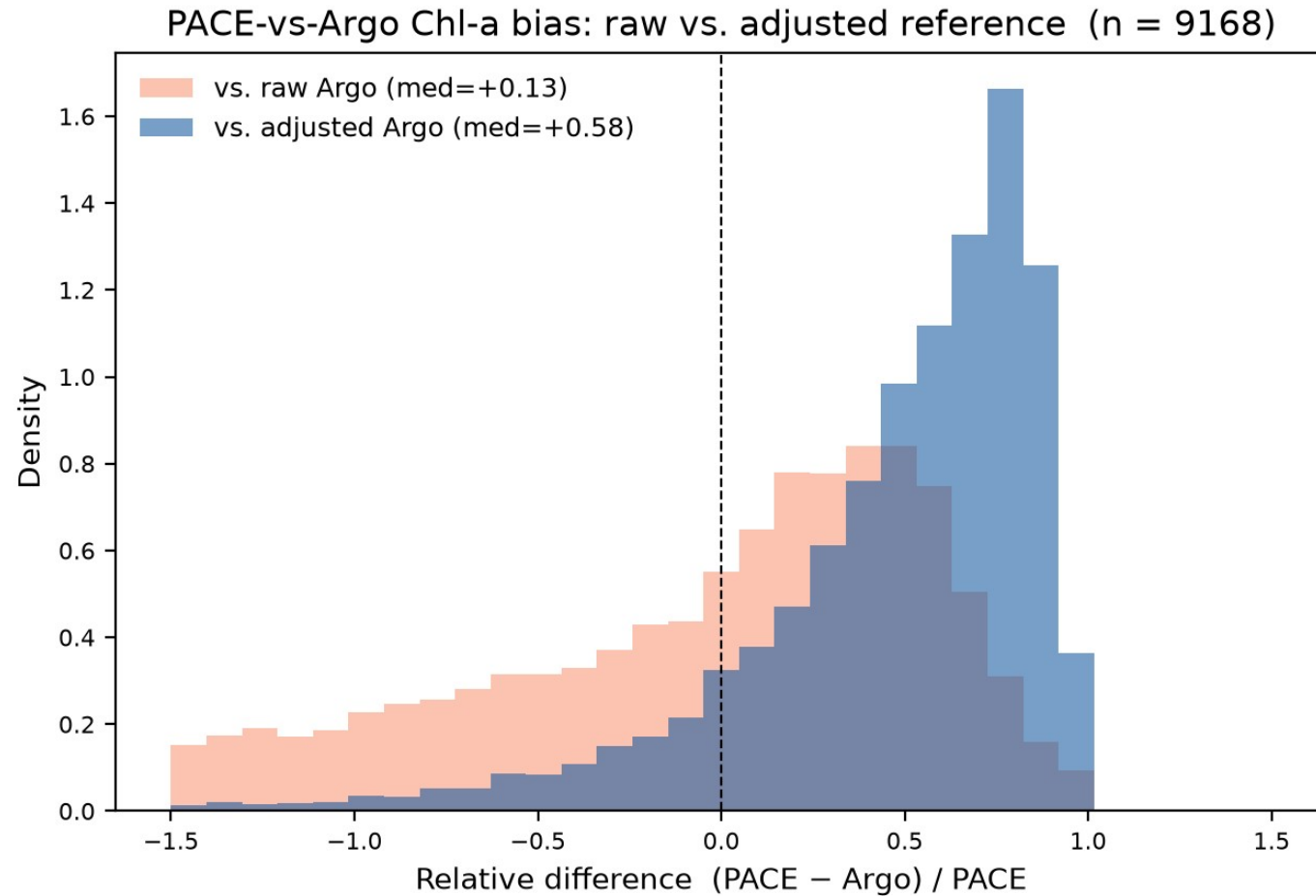
- 58 % PACE > Argo; median +0.13; rank correlation 0.78
- no single bias describes the population — next slide

The chlorophyll bias changes sign with concentration



Octile medians +0.65 in the clearest water to -0.8 in the richest, zero crossing near 0.1–0.2 $mg\ m^{-3}$. Geometry and PACE quality ruled out: the highest-quality subset is more biased (+0.22).

The in-situ reference is itself ambiguous, by a factor of ~4.5



Scoring against delayed-mode-adjusted rather than raw Argo chlorophyll moves the median from +0.13 to +0.58. Ruling: adjusted is the headline, raw the sensitivity band.

What else the matchups say, and what is excluded

- **Ruled out as drivers**

- time separation (+0.05 to +0.35 by octile), distance (+0.10 to +0.15), PACE flags (quality subset +0.22)

- **Structure that remains**

- Southern Ocean -0.53 against +0.14 to +0.35 elsewhere; season +0.15-0.20 (Mar-Oct) vs +0.03-0.09
- AOML-processed floats +0.06 vs +0.25 for all other data centres

- **Mechanism candidate**

- bias vs fitted a_{dg} : $\rho = -0.24$ — dissolved absorption aliased into the phytoplankton term at high a_{dg}

- **CDOM: qualitative by design**

- float measures fluorescence in relative units; BING retrieves an absorption; the fleet's CDOM sensors carry an unapplied 5.6× correction
- rank correlation +0.13 on 3,676 restricted matchups — that is all that is claimed

- **Excluded**

- PACE-vs-Argo backscatter validation is a collaborator's thesis
- A NASA GIOP retrofit over all 14,609 matchups: BING and GIOP agree on sign and share a low-concentration overestimate — the boundary condition taken from it

PAB 2.0: the mission-scale echo of the RT ladder, under way

- **What it is**

- re-fit of every matchup under the inelastic forward model of Chapter 2 (robust_hybrid + Raman + Chl fluorescence)
- per-pixel solar and sensor geometry read from the PACE L1B granule (the L2 product carries none)
- free B_p; version-aware ids; separate 2.0 database
- **Status (2026-09-17)**
- pipeline built; image validated in-cluster; backfill of 1,690 never-attempted profiles planned
- first 60 real fits: 2.0 retrieves 26 % less b_{bp}(700) than 1.0

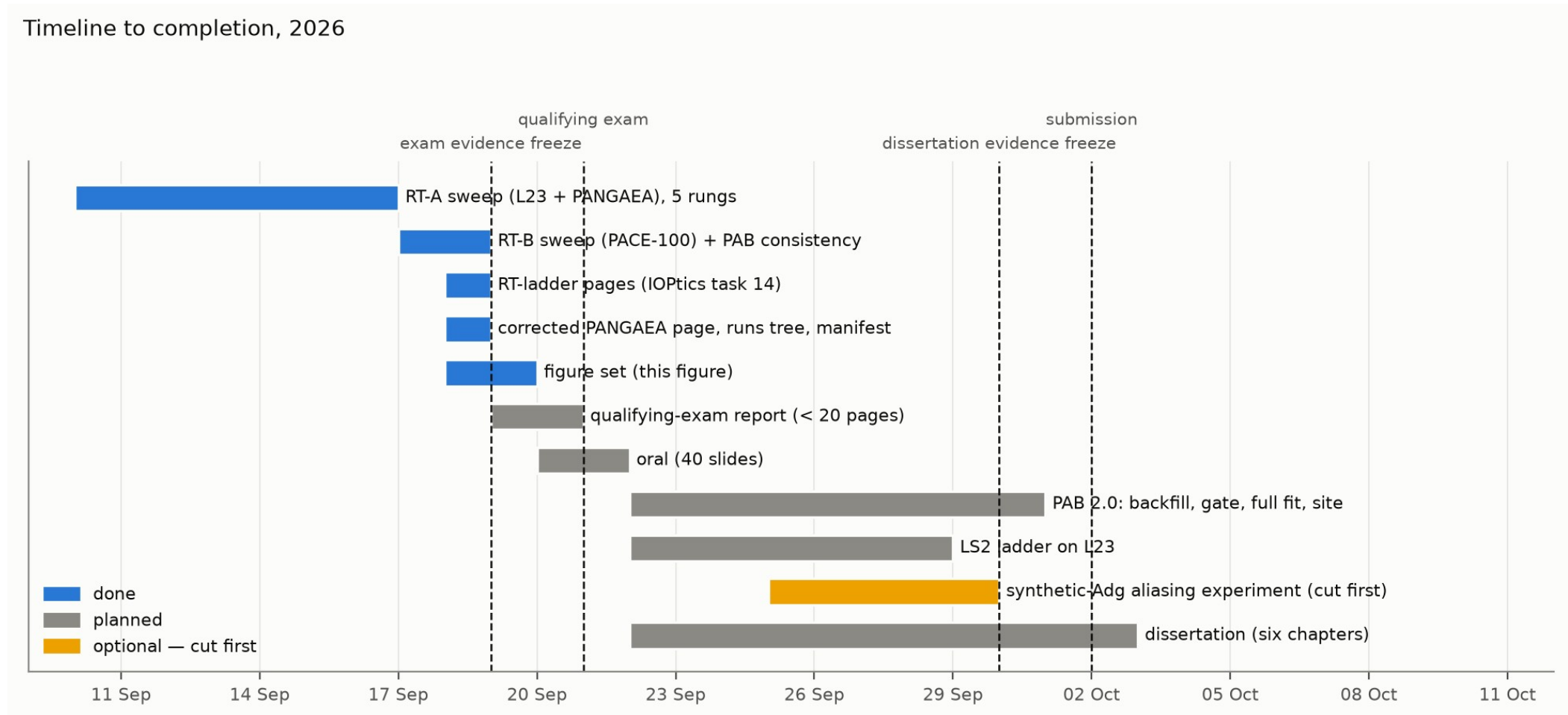
- **What it will answer**

- does the chlorophyll sign change survive the physics? (the claim says yes)
- does mission-scale backscatter move as the ladder predicts? (the claim says yes, by about a fifth)

- **What it will not claim**

- the bb_p(700) comparison is reported in PAB's own run report and offered to the collaborator, not claimed here
- **Caveat carried everywhere**
- population statistics are for the matched set until the backfill runs

Timeline to completion



Everything shown today existed by the 19 September freeze. PAB 2.0, LS2 and the synthetic-`a_dg` test follow before the 30 September dissertation freeze; the last is cut first.

Risks, and what retires each

risk	consequence	what retires it
aliasing hypothesis falsified	PAB mechanism section rewritten; claim stands on L23 + GLORIA	synthetic-a_dg experiment; a null is reported
PAB 2.0 full run misses 09-30	Ch 5 reports the gate and v1.0	backfill + gate the week of 09-22
Argo reference ambiguous (~4.5×)	headline +0.13 or +0.58	ruling: adjusted headline, raw sensitivity
forward model off its zenith grid	bounds hold inside 0–60° only	stated; denser HydroLight zeniths are future work
CDOM fluorescence unvalidated	rung 5 is a plausibility test	flagged; rung 4 is the like-for-like
EPFT-UP retrain blocked (PML)	Ch 4 cannot reproduce training	chapter stands on audit + transfer
1,690 PAB profiles never attempted	rates are for the matched set	the 2.0 backfill; footnoted until then
citations on branches (47 of 62)	main does not yet resolve them	advisor merges after submission; manifest records branch + commit

Conclusions

The claim, restated

Semi-analytical retrieval of ocean inherent optical properties fails in two separable ways. Forward-model (physics) error is identifiable and removable with a sufficiently accurate differentiable radiative-transfer model; once it is removed, the residual error is parameterization error, concentrated in the split of absorption between phytoplankton and CDOM + detritus.

Established today: forward-model error $\rightarrow$ 0.3 %; with the parameterization fixed, physics removes the backscatter error and leaves the absorption split; with the physics fixed, parameterizations move the split by factors and totals hardly at all; the same degeneracy one level up and at mission scale.

Falsifiable, and two tests remain: LS2 should match BING on totals and be unable to touch the split; PAB 2.0 should move backscatter and leave the chlorophyll sign change.

On the candidate: the arrangement produced a complete provenance trail, and its characteristic failure — treating the repository as the world — which the record shows and the advisor corrected.

Backup

After the end

Ordered by the questions this evidence most invites:

1. How do you know the forward model is right?
2. Why is a_{ph} wrong by a factor of two, and should anyone care?
3. What would falsify the claim?
4. What did the candidate do, and what did the advisor do?
5. The ladder numbers, in full.
6. Where every citation lives.

B1 — How do you know the forward model is right?

- **Against what**

- HydroLight, the numerical radiative-transfer solution that generated L23; 9,960 samples (3,320 bodies × 3 zeniths)

- **Held out how**

- 20 % of bodies by scene: 0.30 % rRMS; the 60° zenith when trained on 0/30: 4.7–12.2 % (5 seeds) — reported, not gated

- **Cross-checked**

- inelastic terms match BING's reference implementation to 10^{-13} – 10^{-16} ; gradients to 5×10^{-9}

- **Not shown right**

- phase function (1.7× span in B_p), off-nadir view, CDOM fluorescence, $\lambda < 400$ nm (13 % at 350)

- **Where the ladder stands relative to those edges**

- L23 arm at 0° zenith, inside the grid; PANGAEA/PACE zeniths computed per record, inside 0–60°
- B_p free on L23 and PACE — posteriors near 0.027, above the trained span [0.010, 0.018]: DomainWarning on every hybrid fit, accepted with the caveat; the analytic ZTT rung is the control

- **The honest answer**

- right to 0.3 % inside the trained domain, degrading to ~10 % on unseen sun angle; every result in the talk is inside the domain

B2 — Why is a_{ph} wrong by a factor of two, and should anyone care?

- **Why**
 - a_{ph} and a_{dg} both absorb most in the blue; their sum is set by the reflectance level, their split by shape alone
 - BING's a_{ph} is a fixed Bricaud shape scaled by one amplitude; a_{dg} an exponential with one amplitude and one slope
 - the physics changes what light is available in the red; the parameterization decides where it goes — and it goes into a_{ph}
 - GIOP and GSM under the same physics: 60 % and 28 % — different parameterizations, same disease (slide 23)
- **Should anyone care**
 - a_{ph} is the chlorophyll proxy; PACE's phytoplankton community products are built on the decomposition
 - at mission scale it appears as a chlorophyll bias that changes sign with concentration and tracks a_{dg} (slide 33)
 - one level up it appears as a species algorithm that transfers for one group and not two (slide 29)
- **What would fix it**
 - not physics. Either more information (K_d, as LS2 uses) or a parameterization that admits it cannot split — which is the dissertation's LS2 test

B3 — What would falsify the claim?

- **If completing the physics had fixed a_{ph} / a_{dg}**
 - the claim would be wrong; it did not, on 3,309 synthetic bodies (slide 14)
- **If the LS2 comparison shows a semi-analytical form that recovers the split from R_{rs} alone**
 - the "parameterization" half is wrong; LS2 (Loisel+2018) declines to split and should match BING on totals — the dissertation runs the three-rung ladder (true K_d + true b_p ; true K_d + OC4 b_p ; NN K_d + OC4 b_p)
- **If PAB 2.0 removes the chlorophyll sign change**
 - the mission-scale mechanism is the physics after all; the claim says the sign change survives and backscatter moves by ~20 %

- **If the synthetic- a_{dg} experiment shows no aliasing**
 - the PAB mechanism section is rewritten; the L23 ladder and GLORIA stand on their own
- **If GLORIA turbid spectra are fitted by a richer b_{bp} under the new physics**
 - then the turbid failure was parameterization after all; today four b_{bp} models give one fit and the forward model is the suspect
- **Already retired**
 - "RT-B may not run" — it ran on 2026-09-17

B4 — What did the candidate do, and what did the advisor do?

- **The advisor**

- chose the four problems and the exclusions; wrote every numbered prompt and every Q&A answer (55 questions in the RT tests alone)
- judged acceptance at every gate (design docs, coding stages, sweep launches, the 100-matchup PAB gate)
- ran every git command; owns the machines, the data, the cluster and the Drive
- is the author of BING and the 2025 paper the thesis stands on

- **The candidate (Claude: Opus 4.6–5, Fable 5–5.1, over eight months)**

- wrote the packages: IOptics (12.6k lines, 535 tests), robust.rt (8.3k, 537 tests), PAB (8.9k, 140–208 tests), EPFT-UP; the BING extensions
- designed and ran the sweeps; wrote the reports, the pages, the figures, this deck and the written report
- **The record**
 - ~530 dated session logs since 2026-06-15; every result traces to one; every citation to a path, branch and commit
- **The failure mode the record shows**
 - inferring absence of work from absence of commits — twice; corrected by the advisor and by re-checking the disk

B5 — The L23 ladder in full (MCMC, 3,309 bodies, fractional error / bias)

rung	a(440)	a_ph(440)	a_dg(440)	b_bp(555)	b_bp(670)	χ^2_v
elastic, analytic	5.3 / +2.8	91 / -33	26 / +6	55 / +55	68 / +68	1.12
elastic, hybrid	5.2 / +3.2	135 / -48	29 / +14	41 / +41	60 / +60	1.14
+ Raman	5.1 / +3.6	252 / -69	38 / +31	12 / +10	26 / +26	1.14
+ Chl fluorescence	5.9 / +4.9	85 / -38	26 / +20	10 / +6	18 / +16	1.07
+ CDOM fluorescence	5.0 / +2.8	93 / -40	25 / +17	12 / -5	13 / +6	1.08

Percent. Coverage of the 68 % interval on b_bp(555): 0.00, 0.02, 0.52, 0.63, 0.59 down the ladder. Every cell reconciled against the IOptics page (reports/rta_reconcile.md, 0 discrepancies).

B6 — Where every citation lives

- **62 artefacts cited; 47 not yet on main**
 - RoB inelastic-rt (14), IOptics rt-tests (18), PAB full-inelastic (6) + hyper_matchups (1), BING rob_cdom (3), this repository's qualifying-exam branch (4)
 - EPFT-UP and ocpy: everything on main
- **The manifest records, for each, the path, the branch and the last-touching commit**
 - the advisor has undertaken to make main resolve them after submission
- **Sweep outputs are not in git**
 - parquet + chains under \$OS_COLOR/IOptics/runs on the canonical workstation; each page stamps the code and config that produced it

- **Repositories**
 - github.com/ocean-colour/{IOptics, retrieve-or-bust, PAB, bing, EPFT-UP, ocpy}
 - github.com/Sea-Meets-the-Stars/claudes-phd-thesis
- **Documentation sites**
 - ioptics.readthedocs.io · retrieve-or-bust.readthedocs.io · pab-report.readthedocs.io · epft-up.readthedocs.io
- **DOI**
 - EPFT-UP v0.1.0: 10.5281/zenodo.22797970; others after submission